

PCPOP : A Julia library for polynomial optimization with partially commuting variables

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Abstract

1 Introduction

PCPOP is a Julia library that builds and solves SDP relaxations for polynomial optimization with partially-commuting variables, where some variables commute and some not, and hence generalizing the non-commutative and fully-commutative polynomial optimization. It supports both the eigen-value and tracial versions. The algebra associated with Partially-commutative polynomial optimization is the Partially-commutative polynomial algebra [see [**<empty citation>**]] and it can be seen as special case of Graph product polynomial algebra [see ??]. At the core,

- PCPOP implements various data structures to efficiently build monomials belonging to Graph product algebra and do arithmetic with them.
- PCPOP implements various algorithms, the theory of which is mentioned in [**<empty citation>**], to find normal forms of polynomials under given polynomial identities.

Optimizing a polynomial given some polynomial constraints, a.k.a. polynomial optimization, has become a central tool in many diverse fields, but is a NP-hard problem. The fully-commutative polynomial optimization was approximated by Lasserre using a sequence of semi-definite programs(SDP), the optimal values of which converge to global optimum of the polynomial optimization under certain assumptions. This was then extended to Non-Commuting variables yielding the NPA hierarchy [**<empty citation>**]. Given the natural structure of non-commutativity among operators in quantum physics, the NPA hierarchy finds numerous applications such as solving foundational questions, analyzing Device-Independent Quantum Information protocols, Self-Testing, and many-body physics. It has also been extended to trace polynomials, with many prominent applications, such as the analysis of quantum networks.

Given $p(\underline{x}), q_i(\underline{x})$, polynomials in $\underline{x}=(x_1, x_2 \dots x_n)$ variables, polynomial optimisation can be described as,

$$\begin{aligned} p^* &= \inf_{\underline{x}} p(\underline{x}), \\ s.t \quad q_i(\underline{x}) &\geq 0. \end{aligned} \tag{1}$$

In non-commuting variables $\mathbf{X}=(\mathbf{x}_1, \mathbf{x}_2 \dots \mathbf{x}_n)$ and polynomials $p(\mathbf{X}), q_i(\mathbf{X})$, polynomials in free algebra, non-commuting polynomial optimization can be described as,

$$\begin{aligned} p^* &= \inf_{\underline{A} \in \mathcal{B}(\mathcal{H})^n, \psi \in \mathcal{H}, \mathcal{H}} \langle \psi, p(\underline{A}) \psi \rangle, \\ s.t \quad q_i(\underline{A}) &\geq 0. \end{aligned} \tag{2}$$

In non-commuting variables $\mathbf{X}=(\mathbf{x}_1, \mathbf{x}_2 \dots \mathbf{x}_n)$ and polynomials $p(\mathbf{X})$, $q_i(\mathbf{X})$, polynomials in free algebra, trace polynomial optimization can be described as,

$$\begin{aligned} p^* &= \inf \operatorname{tr}(p(\mathbf{X})), \\ \text{s.t. } & q_i(\mathbf{X}) \geq 0. \\ & \operatorname{tr}(r_j(\mathbf{X})) \geq 0 \end{aligned} \tag{3}$$